Neonatal Hypoxic Ischemic Encephalopathy (HIE)

Diagnosis and Management Guidelines
Version dated 5-Jul-2021

Neonatal HIE commonly presents with encephalopathy but if oxygen deprivation or loss of blood flow (ischemia) to the brain in the perinatal period is suspected, it is just as serious.

Remember: Not all neonatal encephalopathy is HIE!

Cool Cool (www.coolcool) Interactive guide to neonatal care before HIE-HIE that may benefit from cooling

Potential Consultations at T0
- can worsen congenital/early onset/perinatal sepsis
- can worsen pulmonary hypertension
- **pneumothorax or SCALP** (if cooling)
- **Neurology** (if seizures, neonatal hemorrhage or other neurological issue, or if cooling is considered especially if infant is coagulopathic in addition to being encephalopathic)

Inclusion Criteria for Therapeutic Hypothermia (TH) (1-3):
(1) Gestational age of at least 36 weeks
(2) Age < 6 hours at time of enrollment
AND one of the following:

Biochemical Criteria
Blood gas within 1st hr of life
(cord, ABG, VBG, radial, peripheral) OR base deficit >/=16

Alternative Criteria
If no blood gas available OR
base deficit 12-15
pH 7.00-7.15

AND BOTH of the following:

(1) 10 min APGAR >/=5 OR assisted ventilation at birth for >/=10 mins
(2) Acute Perinatal Events (intrapartum hypoxia)

AND
either **Seizures** or **Moderate or Severe Encephalopathy**
requires at least 2 from modifiers in severe category
Grade **Moderate** (Sarnat & Sarnat 1976) or **Severe** (Sarnat & Sarnat 1976)

	Severe Encephalopathy	Moderate Encephalopathy	Mild Encephalopathy
Level of Consciousness	Stupor or coma	Lethargic	Hyperalert, irritable
Spontaneous Activity	No activity	Decreased activity	Slightly decreased activity
Posture	Decerebrate	Distal flexion, complete extension	Mild distal flexion
Tone	Flaccid	Attenuates in time	Hypotonic
Primitive Reflexes	Absent suck, Moro	Weak suck, Moro	Suck is usually normal
	Stare	Dysmetric Moro	Mild with not threatened Moro
Autonomic system	Seizures, always abnormal, frequent	Frequent, common	Normal
	Variable heart Rate, bradycardia	HR normally, irregular modeling	

If criteria met, initiate TH

This includes:
- cooling to 33.5 HE (rectal and esophageal)
- 72 hours of cooling, then slow rewarming (rectal temp 0.5/hr)
- treat ultrasound (rule out large hemorrhage or other cause of encephalopathy, treat if hypotensive)
- data prior to initiating hypothermia(pH, arterial/venous CBC, INR, PTT, CMP, mag, phos, blood gas (arterial/venous) + lactate), POC glucose (with cal), then q6hr
- CXR, abdominal US, cardiac echo, head US, EEG, aEEG
- continuous EKG, continuous pulse ox
- continuous BP monitoring (if hypotensive, consider longer if abnormal renal/ adrenal/ liver or first 24-48 hours or during rewarming)(1)
- Hypotension: treat with volume, inotrope, steroids, pressors as required
- Hypo-fevers may be considered during TH at discretion of provider (recommend against hyperthermia, treat with tylenol, cooling blanket, cooling cap)
- morphine for shivering/comfort, can also consider dexmedetomidine (2)

TH is continued for 72 hours, at which time patient will be rewarmed by 0.5 C per hour until reaches 36.5 C.

If not 36.5 following rewarming (+/-DOL 4)(1-3):
- consider repeat head US to rule out possible brain diffusion sequences are areal restrict from **DOL 3-4** in babies undergoing TH **DOL 3-4** in babies not undergoing TH (3) (Consider repeat head US if clinical deterioration, but NOT after DOL 10 after 1-1-21 scan shows up to avoid overdiagnosing injury)
- if stable, consider MRI (not 1 week get psychiatrists (1)

Patients who do NOT meet criteria for TH
- consider head US (if not done)
- EEG for at least 24 hours, or at least 24 hours after last seizure
- consider MRI
- start HIE FU, 2 yr

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